

- What is GRU ? – **Gated Recurrent Unit**

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invented → 2014 by K. Cho. (LSTM was created by 1997)

- This is having the same gene as LSTM Yes, they are very closely related.
- When we say, they are related, they share many good features with the newer generation getting better over the previous one.
- GRUs can effectively address the vanishing gradient problem just like LSTM.
- In addition to it, GRU appears simpler comparing LSTM.

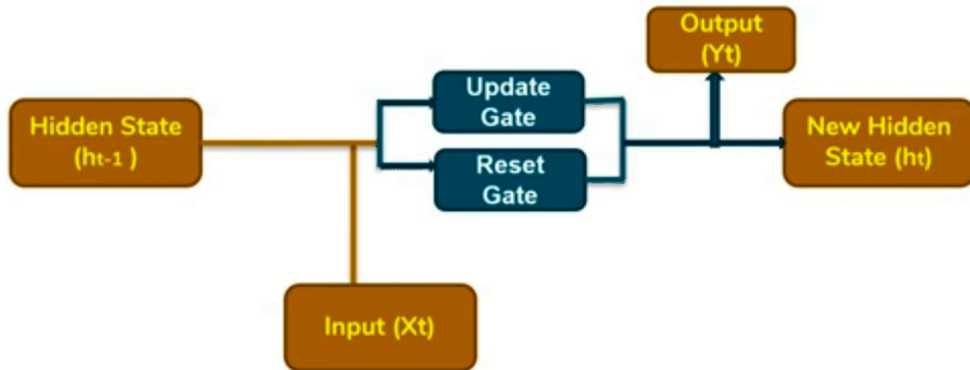


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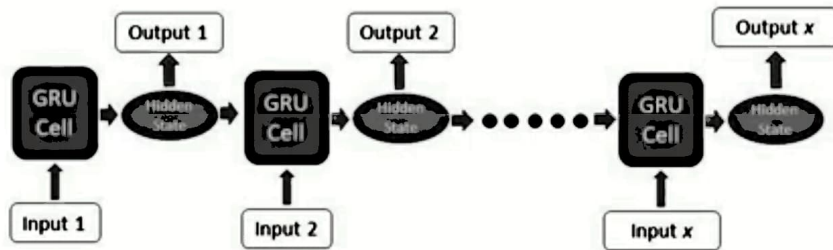
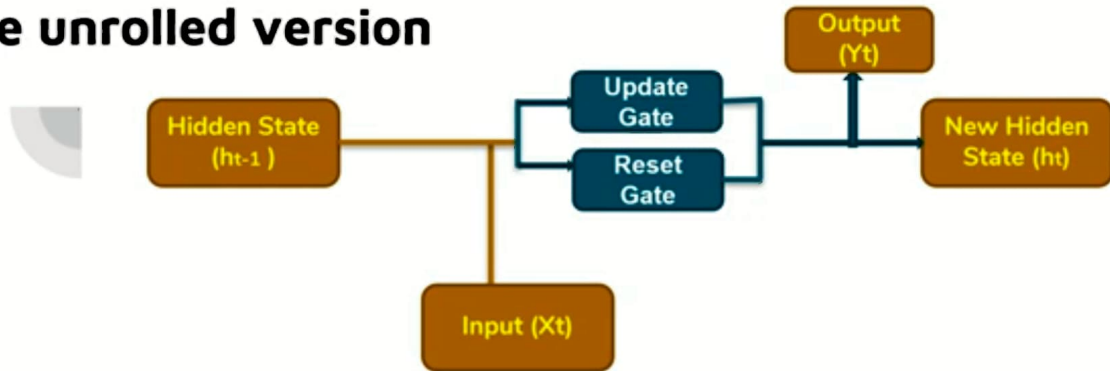
- As we have seen in the LSTM, GRU also utilizing the gating mechanisms (Remember the gates used in LSTM – Forget, Input and Output) to manage and as well to control the flow of the information between the cells in the neural network.
- Can we understand the differences between LSTM and GRU in the architectural perspective as well as the functioning?
- Number of gates in LSTM – 3
- Number of gates in GRU – 2
- GRUs do not have the Cell State and Hidden states are used!
- The gates are named as **Reset** Gate and **Update** Gate.

Contd.,

- The idea is pretty much similar. The gates help in determining what to be retained/passed or dropped.
- The gates shall get the value between 0 and 1 as we have seen earlier. A 0 from the gate is revealing that the data is unimportant. 1 says it is important (It is like closer to 0, it is unimportant, closer to 1 it is important).
- GRUs do not have the Cell State and Hidden states are used!

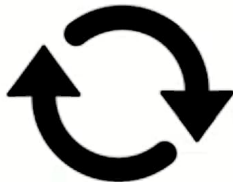


The unrolled version



Contd.,

- Update Gate (z)
- The major task of this gate is to tell the model about "How much of the past information needs to be maintained i.e. to be passed along to the future."
- Reset Gate (r) -
- This gate is used from the model to decide how much of the past information to forget
- As usual there are weights associated with every stage! Remember it!



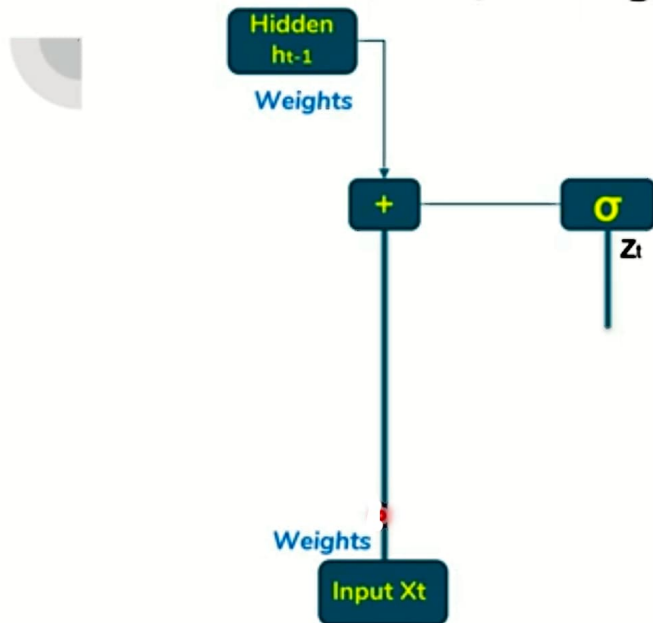


Can we go to math and pictorial representation to understand the functioning?

- Update gate:
- Represented by Z_t
- Input represented by X_t , Previous h_{t-1} State information multiplied with respective weights as the parameters. (Notations differ, nothing to worry as long as you understand that respective weights are to be used appropriately)
- Sigmoid activation is used to derive Z_t .


$$z_t = \sigma(W^{(z)}x_t + U^{(z)}h_{t-1})$$

Can we visualize the update gate ??



$$z_t = \sigma(W^{(z)}x_t + U^{(z)}h_{t-1})$$

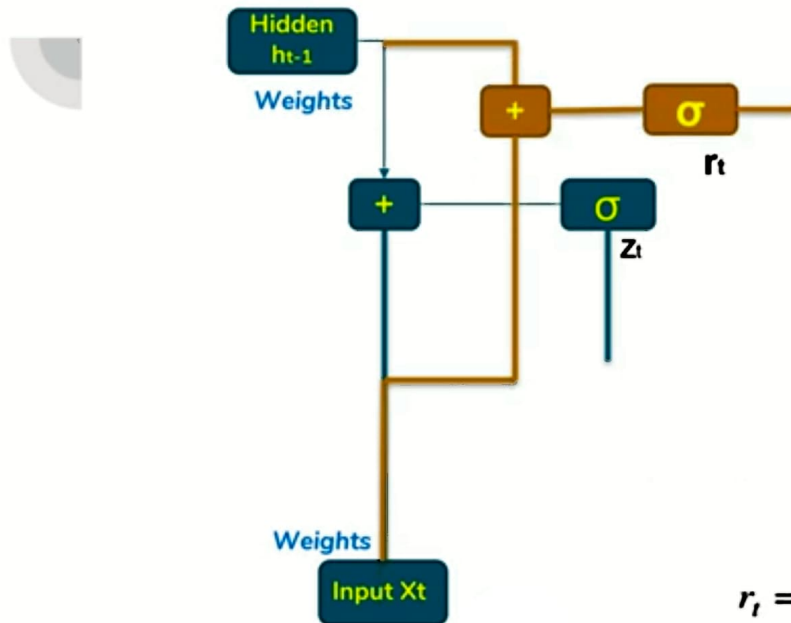


Can we go to math and pictorial representation to understand the functioning?

- Reset gate:
- Represented by $\mathbf{r}_t$
- Input represented by $\mathbf{x}_t$, Previous $\mathbf{h}_{t-1}$ State information multiplied with respective weights as the parameters. (Notations differ, nothing to worry as long as you understand that respective weights are to be used appropriately)
- Sigmoid activation is used to derive $\mathbf{r}_t$.

$$r_t = \sigma(W^{(r)}x_t + U^{(r)}h_{t-1})$$

We can go with the Reset Gate now



$$r_t = \sigma(W^{(r)}x_t + U^{(r)}h_{t-1})$$

Lets get better ..



A New Component is getting introduced now!! 😊 Yes, it is a memory component and will use the reset gate to get the relevant information stored from the past.

You are reviewing a movie. Initially you start with “ the movie is directed by X, it has featured Y, Music by Z, etc.. After about 10 lines, you say, the movie, I think is a bad one for the money I paid”.

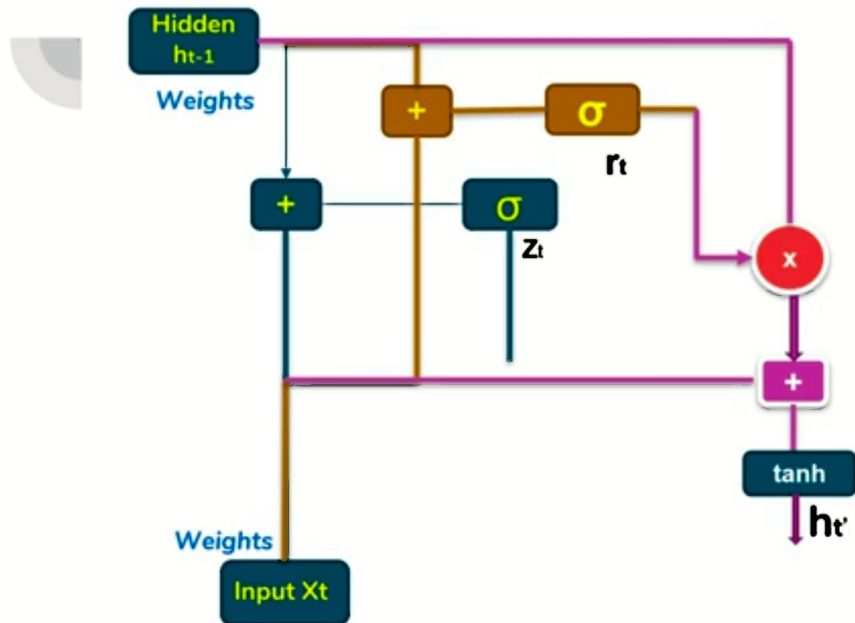
So, which is the actual review? The final line. So, the neural network should not remember the past sentences and focus on the last sentence to get the crux. It is what is enabled here.

Here, r_t will be assigned 0 until the final phrase is in analyze so as to drop the unfocussed items.

Then comes the tanh activation and you get h_t'

$$h_t' = \tanh(Wx_t + r_t \odot Uh_{t-1})$$

Can we visualize



$$h_t = \tanh(Wx_t + r_t \odot Uh_{t-1})$$



We are nearing the end..

- As the last step, the network needs to calculate h_t — vector which holds information for the current unit and passes it down to the network.
- In order to do that the update gate is needed.

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot h'_t$$

